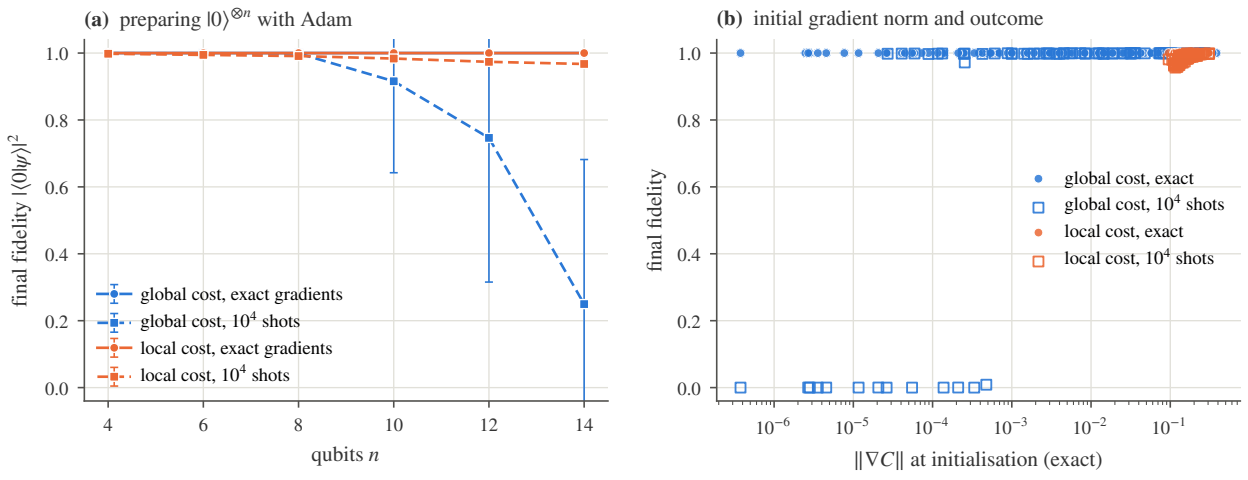
(a) preparing $|0\rangle^{\otimes n}$ with Adam
final fidelity $|\langle 0|\psi\rangle|^2$
1.0
0.8
0.6
0.4
0.2
0.0
4
6
8
10
12
14
qubits n
global cost, exact gradients
global cost, 10^4 shots
local cost, exact gradients
local cost, 10^4 shots
(b) initial gradient norm and outcome
final fidelity
1.0
0.8
0.6
0.4
0.2
0.0
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
$\|\nabla C\|$ at initialisation (exact)
global cost, exact
global cost, 10^4 shots
local cost, exact
local cost, 10^4 shots